# AI APPLICATIONS AND PEAS ANALYSIS

A Study of AI-Based Smart Agriculture and AI in Education

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# 1. Introduction

Artificial Intelligence (AI) is a technology that enables computer systems to perform tasks that normally require human intelligence, such as recognizing patterns, making decisions and learning from data. Machine Learning (ML) is a major part of AI in which systems learn from examples and improve their performance without being explicitly programmed for every situation. In this report, two practical applications of AI are studied: Smart Agriculture and AI in Education. Both applications are analysed using the PEAS framework, which describes an intelligent agent in terms of its Performance Measure, Environment, Actuators and Sensors. Along with the technical aspects, the report also discusses the social, ethical and legal issues that should be considered when AI is used in these areas.

# 2. Types of Artificial Intelligence

AI is usually grouped into three categories based on how capable it is:

## 2.1 Narrow AI / Weak AI

- Designed to perform a specific task or a limited set of tasks.
- Examples include image recognition, recommendation systems and voice assistants.
- Most AI systems used today are Narrow AI.

## 2.2 General AI / Strong AI

- A theoretical type of AI that could perform different tasks like a human.
- It would be able to learn and apply knowledge across different areas.
- General AI has not been achieved yet.

## 2.3 Super AI

- A theoretical form of AI that would be more intelligent than humans.
- It could perform tasks such as advanced reasoning, problem-solving and creativity.
- Super AI does not currently exist.

## 2.4 Where the Two Applications Fit

- Smart Agriculture and AI in Education are examples of Narrow AI.
- A crop disease detection system is designed for detecting diseases and cannot perform unrelated tasks.
- Similarly, an AI tutoring system works within a specific subject or learning task.
- Therefore, both applications are based on Narrow AI, not General AI or Super AI.

# 3. AI-Based Smart Agriculture

## 3.1 Introduction

- Smart Agriculture means using technologies such as AI, sensors and data analysis to improve farming and decision-making. AI can help farmers monitor crops, detect diseases, manage irrigation, identify pests and predict crop yield.
- Precision Agriculture is an important part of Smart Agriculture where resources such as water and fertilizers are used according to the specific needs of crops and soil. AI can analyse information from sensors, cameras, weather data and other sources to provide useful recommendations to farmers.

### 3.1.1 Practical Applications

- Crop disease detection — ML models read leaf images and catch disease symptoms early.
- Crop health monitoring — drone or satellite imagery shows which zones of a field are stressed.
- Soil analysis — sensors and models estimate nutrient and moisture levels to guide fertilizer and irrigation calls.
- Smart irrigation — combines soil-moisture data with weather forecasts to water only when needed.
- Weather-based decision support — advises on sowing, harvesting or spraying windows.
- Pest detection — image or trap-based models identify pest type and severity for targeted treatment.
- Yield prediction — forecasts the harvest using historical yield, soil and weather data.
- Fertilizer recommendation — suggests type and quantity based on soil test and crop stage.
- Precision farming — applies water, fertilizer or pesticide at a variable rate, zone by zone.
- Farm advisory apps/chatbots — turn all this data into simple, actionable advice for the farmer.

## 3.2 Five Intelligent Agents

| Agent No. | Intelligent Agent | Main Function |
|---|---|---|
| A1 | Crop Disease Detection Agent | Analyses crop images to identify possible diseases and alerts the farmer. |
| A2 | Smart Irrigation Agent | Uses soil moisture and weather data to decide when irrigation is required. |
| A3 | Crop Health Monitoring Agent | Monitors crop growth and identifies unhealthy areas.. |

| Agent No. | Intelligent Agent | Main Function |
|---|---|---|
| A4 | Yield Prediction and Farm Advisory Agent | Analyses crop and weather data to estimate yield and provide farming recommendations. |
| A5 | Pest Detection and Management Agent | Detects possible pest attacks and suggests suitable control measures. |

## 3.3 Five Environments

| Environment No. | Environment | Description |
|---|---|---|
| E1 | Crop Field | Open farming area where crops are monitored and managed. |
| E2 | Greenhouse | Controlled environment used for growing crops under monitored conditions. |
| E3 | Irrigation Area | Area where water supply and irrigation are controlled. |
| E4 | Agricultural Storage Area | Place where harvested crops are stored and monitored. |
| E5 | Remote / IoT-Enabled Farm | Farm where sensors and connected devices collect data remotely. |

## 3.4 PEAS Analysis — Ten Combinations

Each combination below pairs one of the five agents with a relevant environment, along with its performance measures, actuators and sensors. Sensors and actuators are chosen to genuinely fit each environment rather than being repeated identically.

## Combination 1

### Crop Disease Detection Agent → Crop Field

**Performance Measure:**

- Early and correct disease identification
- Reduction in crop loss
- Timely alert to farmer

**Environment:**

- Open farmland
- Sowing-to-harvest growing area
- Outdoor field conditions

**Actuators:**

- Mobile/app alert to farmer
- Recommendation of pesticide or treatment

**Sensors:**

- Camera (leaf image capture)

- GPS location sensor
- Humidity sensor

**Description:**
In an open field, the agent captures leaf images through a farmer's phone camera or a field camera, compares patterns against a trained model, and immediately notifies the farmer if a disease is suspected. Quick detection here directly limits how far a disease can spread.

## Combination 2

**Crop Disease Detection Agent → Greenhouse**

**Performance Measure:**
- Detection accuracy under controlled light
- Reduced use of chemical treatment
- Plant survival rate

**Environment:**
- Controlled indoor growing structure
- Regulated temperature and humidity

**Actuators:**
- Automated alert to greenhouse operator
- Trigger for localized treatment sprayer

**Sensors:**
- Fixed overhead camera
- Temperature and humidity sensor

**Description:**
Inside a greenhouse the lighting is more consistent, so the agent can rely on fixed cameras for continuous monitoring rather than one-time photos, allowing it to catch early symptoms before they spread to nearby plants.

## Combination 3

**Smart Irrigation Agent → Crop Field**

**Performance Measure:**
- Water-use efficiency
- Correct irrigation timing
- Reduction in water wastage

**Environment:**
- Open farmland
- Sowing-to-harvest growing area
- Outdoor field conditions

**Actuators:**

- Water pump switch
- Irrigation valve control
- Sprinkler activation

**Sensors:**

- Soil moisture sensor
- Temperature sensor
- Weather forecast feed

**Description:**
The agent checks soil moisture and upcoming weather before deciding whether to irrigate, so water is used only when the crop genuinely needs it, which helps a farmer avoid both under-watering and overwatering.

### Combination 4

**Smart Irrigation Agent → Irrigation Area**

**Performance Measure:**

- Even water distribution
- Pipeline efficiency
- Leakage minimisation

**Environment:**

- Canal and pipeline network
- Valve and pump stations

**Actuators:**

- Valve and pump network control
- Flow rate adjustment

**Sensors:**

- Flow rate sensor
- Pressure sensor
- Soil moisture sensor (zone-wise)

**Description:**
Here the agent manages a whole network of valves and pumps rather than a single field, balancing flow across different zones so that every part of the irrigation area receives water proportionate to its need.

### Combination 5

**Crop Health Monitoring Agent → Crop Field**

**Performance Measure:**

- Accuracy of stress detection
- Coverage of the field
- Response time to alerts

**Environment:**

- Open farmland
- Sowing-to-harvest growing area
- Outdoor field conditions

**Actuators:**

- Generate health map for farmer
- Highlight stressed zones on app

**Sensors:**

- Drone/aerial camera
- NDVI (vegetation index) sensor
- Soil sensor

**Description:**
Using drone imagery, the agent builds a vegetation-health map of the field so the farmer can see at a glance which zones are healthy and which need attention, instead of having to walk the entire field manually.

### Combination 6

**Crop Health Monitoring Agent → Remote / IoT-Enabled Farm**

**Performance Measure:**

- Data transmission reliability
- Timeliness of remote alerts
- Reduction in manual inspection

**Environment:**

- Large or dispersed farmland
- IoT-connected sensor network

**Actuators:**

- Send remote notification to farmer's device
- Log data to cloud dashboard

**Sensors:**

- IoT field sensors
- Satellite imagery feed
- Network connectivity sensor

**Description:**
On a large or scattered farm the agent depends on IoT sensors and satellite data transmitted over a network, letting a farmer who may not be physically present still monitor crop condition from a distance.

## Combination 7

**Yield Prediction and Farm Advisory Agent → Crop Field**

**Performance Measure:**

- Prediction accuracy
- Usefulness of advisory suggestions
- Improvement in final yield

**Environment:**

- Open farmland
- Sowing-to-harvest growing area
- Outdoor field conditions

**Actuators:**

- Generate yield forecast report
- Suggest fertilizer or sowing adjustments

**Sensors:**

- Soil quality sensor
- Historical yield database
- Weather data feed

**Description:**
By combining current field conditions with past yield records, the agent estimates the likely harvest and suggests practical adjustments, such as fertilizer timing, while the crop is still growing.

## Combination 8

**Yield Prediction and Farm Advisory Agent → Agricultural Storage Area**

**Performance Measure:**

- Accuracy of post-harvest yield record
- Storage planning efficiency
- Spoilage reduction

**Environment:**

- Warehouse or silo
- Post-harvest storage zone

**Actuators:**

- Update inventory/storage records
- Alert on storage capacity or spoilage risk

**Sensors:**

- Storage weight/quantity sensor
- Temperature and humidity sensor (storage)

**Description:**

After harvest, the agent shifts to tracking actual quantities coming into storage against its earlier prediction, and also watches storage conditions so produce is not lost to poor temperature or humidity control.

## Combination 9

**Pest Detection and Management Agent → Crop Field**

**Performance Measure:**

- Pest identification accuracy
- Reduction in crop damage
- Timeliness of intervention

**Environment:**

- Open farmland
- Sowing-to-harvest growing area
- Outdoor field conditions

**Actuators:**

- Alert farmer with pest type and location
- Recommend targeted pesticide dosage

**Sensors:**

- Pest trap sensor
- Field camera
- Weather sensor (pest-favourable conditions)

**Description:**
The agent uses trap data and images from the field to recognise which pest is present and how severe the infestation is, then recommends a targeted response rather than blanket pesticide use.

## Combination 10

**Pest Detection and Management Agent → Greenhouse**

**Performance Measure:**

- Early detection rate
- Reduced chemical usage indoors
- Plant protection

**Environment:**

- Controlled indoor growing structure
- Regulated temperature and humidity

**Actuators:**

- Trigger localized pest-control mechanism
- Notify greenhouse staff

**Sensors:**

- Sticky trap sensor
- Indoor camera

**Description:**
In a greenhouse, pest outbreaks can spread quickly through a confined space, so the agent keeps a closer, continuous watch using indoor traps and cameras to catch an infestation while it is still small.

## 3.5 PEAS Summary Table

| No. | Agent | Environment | Main Sensors | Main Actuators |
|---|---|---|---|---|
| 1 | Crop Disease Detection Agent | Crop Field | Camera (leaf image capture) | Mobile/app alert to farmer |
| 2 | Crop Disease Detection Agent | Greenhouse | Fixed overhead camera | Automated alert to greenhouse operator |
| 3 | Smart Irrigation Agent | Crop Field | Soil moisture sensor | Water pump switch |
| 4 | Smart Irrigation Agent | Irrigation Area | Flow rate sensor | Valve and pump network control |
| 5 | Crop Health Monitoring Agent | Crop Field | Drone/aerial camera | Generate health map for farmer |
| 6 | Crop Health Monitoring Agent | Remote / IoT-Enabled Farm | IoT field sensors | Send remote notification to farmer's device |
| 7 | Yield Prediction and Farm Advisory Agent | Crop Field | Soil quality sensor | Generate yield forecast report |
| 8 | Yield Prediction and Farm Advisory Agent | Agricultural Storage Area | Storage weight/quantity sensor | Update inventory/storage records |
| 9 | Pest Detection and Management Agent | Crop Field | Pest trap sensor | Alert farmer with pest type and location |
| 10 | Pest Detection and Management Agent | Greenhouse | Sticky trap sensor | Trigger localized pest-control mechanism |

## 3.6 Societal Impact of AI in Smart Agriculture

### Positive Impacts

- Improved Crop Production: AI helps farmers make better decisions about crops and resources.
- Water Conservation: Smart irrigation can reduce unnecessary water usage.
- Early Disease Detection: AI can identify crop diseases at an early stage.
- Better Resource Management: AI helps in managing water, fertilizers and other resources efficiently.

- Support for Farmers: AI provides useful information and recommendations for farming decisions.

**Negative Impacts**

- High Cost: AI-based equipment and systems may be expensive for small farmers.
- Technology Dependence: Farmers may become dependent on AI systems for important decisions.
- Data Privacy: Farm and crop data may be collected and stored by technology providers.
- Digital Divide: Farmers with limited internet or technical knowledge may not benefit equally.
- Incorrect Decisions: Wrong AI predictions or recommendations may cause financial or crop losses.

## 3.7 Ethical Concerns

- Privacy: Farmers' and farm data should be collected and used responsibly.
- Bias: AI may give unfair or inaccurate results if its training data is biased.
- Transparency: Farmers should understand the purpose and limitations of AI recommendations.
- Accountability: There should be clear responsibility if an AI recommendation causes harm.
- Human Oversight: Important farming decisions should not depend completely on AI.

## 3.8 Legal Issues

- Data Privacy: Farmers' and farm data should be protected from unauthorized use.
- Data Ownership: There should be clear rules about who owns the data collected from farms.
- Liability: Responsibility should be clear if an AI recommendation causes crop or financial loss.
- Cybersecurity: AI and IoT systems should be protected from hacking and unauthorized access.
- Intellectual Property: AI-generated recommendations, software and agricultural data may involve ownership and copyright issues.

## 3.9 Conclusion

AI can support agriculture by improving crop monitoring, irrigation, disease detection and farm decision-making. However, issues such as data privacy, cost, bias and incorrect recommendations should also be considered. AI should support farmers in decision-making rather than completely replace human judgement.

# 4. AI in Education

## 4.1 Introduction

AI in Education refers to the use of Artificial Intelligence and Machine Learning to support teaching and learning. AI can analyse student performance, provide personalized learning materials, give feedback and assist teachers with routine tasks. It can also help identify areas where students may need additional support. However, AI should be used as a support tool and not as a complete replacement for teachers.

### 4.1.1 Practical Applications

- Personalized learning — content and pace adjusted to each student's level.
- Automated assessment — quizzes and assignments graded automatically with instant feedback.
- Intelligent tutoring — step-by-step guidance based on where a student is stuck.
- Performance prediction — past data flags students who may need extra support.
- Learning material recommendation — relevant videos, readings or exercises based on history.
- AI educational assistants — chatbots answer routine queries and share reminders.
- Attendance and engagement analysis — participation patterns flag disengaged students.
- Early identification of at-risk students — combined performance and engagement data catch problems before results drop sharply.

## 4.2 Five Intelligent Agents

| Agent No. | Intelligent Agent | Main Function |
|---|---|---|
| A1 | Intelligent Tutoring Agent | Teaches a topic step by step, adapts explanations to the learner's pace and gives hints when the student is stuck. |
| A2 | Student Performance Analysis Agent | Studies quiz scores, assignment marks and activity logs to identify students who may be falling behind. |
| A3 | Personalized Learning Recommendation Agent | Suggests the next topic, video or exercise for a student based on their past performance and learning pattern. |
| A4 | Automated Assessment Agent | Evaluates answers, assignments or quizzes automatically and generates marks along with feedback. |
| A5 | AI Educational Assistant | Answers routine student queries, reminds about deadlines and guides students to the right resource or teacher. |

## 4.3 Five Environments

| Environment No. | Environment | Description |
|---|---|---|
| E1 | Classroom | A physical or smart classroom where a teacher conducts lessons, often supported by digital boards or apps. |
| E2 | Online Learning Platform | A web-based LMS or MOOC platform where students access lectures, quizzes and course material. |
| E3 | Computer Laboratory | A lab session where students practise coding or software tasks under supervision. |
| E4 | Examination / Evaluation Environment | An online or proctored test setting where student responses are collected and evaluated. |
| E5 | Student Home / Remote Learning Environment | The student's home or any remote location used for self-study or attending online classes. |

### 4.4 PEAS Analysis — Ten Combinations

As with the agriculture section, each combination below matches an agent to a relevant learning environment, using appropriate digital or data-based sensors and actuators where physical sensors do not apply.

## Combination 1

**Intelligent Tutoring Agent → Online Learning Platform**

**Performance Measure:**

- Learning improvement
- Response accuracy
- Student engagement
- Course completion

**Environment:**

- LMS / MOOC platform
- Web-based course environment

**Actuators:**

- Give explanations
- Provide hints and questions
- Recommend exercises

**Sensors:**

- Student answers
- Quiz scores
- Time spent on topics

**Description:**

On an online platform the agent watches how a student answers practice questions and how long they take, then adjusts the difficulty and style of its explanations so the student is neither bored nor overwhelmed.

## Combination 2

**Intelligent Tutoring Agent → Student Home / Remote Learning Environment**

**Performance Measure:**

- Self-study effectiveness
- Concept retention
- Consistency of practice

**Environment:**

- Student's home
- Remote self-study setting

**Actuators:**

- Send practice reminders
- Provide step-by-step problem walkthroughs

**Sensors:**

- Login and activity pattern
- Past quiz history

**Description:**
Without a teacher physically present, the agent takes on a larger guiding role at home, reminding the student to practise regularly and walking them through problems they get stuck on.

## Combination 3

**Student Performance Analysis Agent → Classroom**

**Performance Measure:**

- Accuracy of at-risk identification
- Improvement in class average
- Timeliness of teacher alerts

**Environment:**

- Physical or smart classroom
- Teacher-led session

**Actuators:**

- Flag struggling students to teacher
- Generate performance summary report

**Sensors:**

- Attendance record

- In-class test scores
- Assignment submission data

**Description:**
The agent tracks each student's classroom performance over time and notifies the teacher when a student's scores or attendance start dropping, so the teacher can step in early.

## Combination 4

**Student Performance Analysis Agent → Examination / Evaluation Environment**

**Performance Measure:**
- Prediction reliability
- Fairness of analysis
- Usefulness of the generated report

**Environment:**
- Online or proctored test setting
- Assessment platform

**Actuators:**
- Generate performance analytics dashboard
- Highlight weak topic areas

**Sensors:**
- Exam scores
- Question-wise response data
- Time taken per question

**Description:**
During evaluation, the agent studies which questions or topics a student struggled with, producing a topic-wise breakdown that helps both the student and teacher target revision effectively.

## Combination 5

**Personalized Learning Recommendation Agent → Online Learning Platform**

**Performance Measure:**
- Relevance of recommendations
- Increase in course engagement
- Learning path efficiency

**Environment:**
- LMS / MOOC platform
- Web-based course environment

**Actuators:**
- Recommend next lesson or video

- Adjust learning path

**Sensors:**

- Learning history
- Content interaction data
- Quiz performance

**Description:**
Based on what a student has already completed and how well they performed, the agent suggests the most suitable next resource, avoiding topics the student has already mastered and reinforcing weak ones.

## Combination 6

**Personalized Learning Recommendation Agent → Computer Laboratory**

**Performance Measure:**

- Task-appropriateness of recommendations
- Skill improvement rate

**Environment:**

- Practical or coding lab session
- Supervised lab environment

**Actuators:**

- Suggest next practical exercise
- Recommend reference material

**Sensors:**

- Code submission history
- Error/debug logs

**Description:**
In a lab setting the agent looks at the kind of coding errors a student repeatedly makes and recommends practice exercises that target exactly those gaps rather than generic assignments.

## Combination 7

**Automated Assessment Agent → Examination / Evaluation Environment**

**Performance Measure:**

- Grading accuracy
- Consistency across students
- Turnaround time

**Environment:**

- Online or proctored test setting
- Assessment platform

**Actuators:**

- Generate marks
- Provide written feedback

**Sensors:**

- Submitted answer text/code
- Answer key/rubric data

**Description:**
The agent compares a student's submitted answer against a rubric or expected answer pattern to assign marks quickly and consistently, which is especially useful for large classes.

## Combination 8

**Automated Assessment Agent → Online Learning Platform**

**Performance Measure:**

- Feedback usefulness
- Grading speed
- Reduction in teacher workload

**Environment:**

- LMS / MOOC platform
- Web-based course environment

**Actuators:**

- Auto-grade quizzes
- Release instant feedback to student

**Sensors:**

- Quiz responses
- Assignment upload data

**Description:**
For platform-based quizzes the agent grades objective and short-answer responses instantly, letting students see their result and feedback right after submission instead of waiting for manual checking.

## Combination 9

**AI Educational Assistant → Classroom**

**Performance Measure:**

- Query resolution accuracy
- Response time
- Student satisfaction

**Environment:**

- Physical or smart classroom
- Teacher-led session

**Actuators:**

- Answer student queries
- Direct student to teacher when needed

**Sensors:**

- Student text/voice query
- Classroom schedule data

**Description:**
Acting like a helpful classroom aide, the agent answers simple, repetitive questions immediately and escalates anything it cannot handle to the teacher, saving classroom time.

## Combination 10

**AI Educational Assistant → Student Home / Remote Learning Environment**

**Performance Measure:**

- Availability
- Accuracy of guidance
- Reduction in missed deadlines

**Environment:**

- Student's home
- Remote self-study setting

**Actuators:**

- Send deadline reminders
- Answer course-related queries

**Sensors:**

- Student query input
- Course calendar data

**Description:**
At home, where a teacher is not immediately reachable, the assistant becomes the first point of contact for a student's doubts and helps them stay on track with assignment deadlines.

## 4.5 PEAS Summary Table

| No. | Agent | Environment | Main Sensors | Main Actuators |
|---|---|---|---|---|
| 1 | Intelligent Tutoring Agent | Online Learning Platform | Student answers | Give explanations |
| 2 | Intelligent Tutoring Agent | Student Home / Remote Learning Environment | Login and activity pattern | Send practice reminders |
| 3 | Student Performance Analysis Agent | Classroom | Attendance record | Flag struggling students to teacher |

| No. | Agent | Environment | Main Sensors | Main Actuators |
|---|---|---|---|---|
| 4 | Student Performance Analysis Agent | Examination / Evaluation Environment | Exam scores | Generate performance analytics dashboard |
| 5 | Personalized Learning Recommendation Agent | Online Learning Platform | Learning history | Recommend next lesson or video |
| 6 | Personalized Learning Recommendation Agent | Computer Laboratory | Code submission history | Suggest next practical exercise |
| 7 | Automated Assessment Agent | Examination / Evaluation Environment | Submitted answer text/code | Generate marks |
| 8 | Automated Assessment Agent | Online Learning Platform | Quiz responses | Auto-grade quizzes |
| 9 | AI Educational Assistant | Classroom | Student text/voice query | Answer student queries |
| 10 | AI Educational Assistant | Student Home / Remote Learning Environment | Student query input | Send deadline reminders |

## 4.6 Societal Impact

### Positive Impacts

- Personalized Education: AI can provide learning material according to a student's pace and level.
- Improved Accessibility: Students, including those learning remotely, can access learning resources more easily.
- Faster Feedback: AI can provide quick feedback on quizzes and other assessments.
- Reduced Teacher Workload: AI can help with repetitive tasks such as basic grading and answering common questions.
- Support Outside Class: AI tools can provide learning assistance beyond regular classroom hours.
- Early Identification of Learning Difficulties: AI can help identify students who may need additional academic support.
- More Learning Resources: Recommendation systems can suggest useful courses, topics and study materials.

### Negative Impacts

- Job Concerns: Teachers may be concerned about automation of some educational tasks.
- Overdependence on AI: Excessive use of AI may reduce independent learning and problem-solving.

- Digital Divide: Students without reliable devices or internet access may not benefit equally.
- Reduced Human Interaction: Overuse of AI may reduce interaction between students and teachers.
- Incorrect Recommendations: AI may give unsuitable recommendations if it incorrectly evaluates a student's ability.
- Student Privacy: Continuous collection of learning data can create privacy concerns.
- Academic Dishonesty: Students may misuse generative AI for assignments and other academic work.

## 4.7 Ethical Concerns

- Privacy: Student data should be collected and used responsibly.
- Surveillance: Continuous monitoring of student activity can affect privacy.
- Bias: Biased data may lead to unfair results or recommendations.
- Transparency: Students and teachers should know how AI is being used.
- Accountability: There should be clear responsibility for incorrect AI decisions.
- Academic Integrity: AI should not be misused for cheating or submitting work without genuine understanding.
- Human Oversight: Teachers should remain involved in important academic decisions.
- Overdependence on AI: Excessive use of AI may affect students' independent thinking and problem-solving skills.

## 4.8 Legal Issues

- Data Privacy: Student information should be protected from unauthorized use.
- Data Protection: Educational institutions should store and handle student data securely.
- Consent: Students should be informed about how their data is collected and used.
- Copyright: AI-generated or AI-assisted content may create copyright and ownership issues.
- Liability: Responsibility should be clear if an AI system provides an incorrect or harmful decision.
- Cybersecurity: AI-based educational systems should be protected from hacking and unauthorized access.

## 4.9 Conclusion

AI can support education through personalized learning, automated assessment and academic assistance. It can reduce repetitive work for teachers and provide additional support to students. However, issues such as privacy, bias, academic integrity and overdependence on AI should be considered. AI should support teachers and students while keeping human judgement involved in important educational decisions.

# 5. Comparison of Both AI Applications

| Parameter | Smart Agriculture | AI in Education |
|---|---|---|
| Main Objective | Support farming decisions using data | Support teaching and learning using data |
| Main AI Techniques | Computer vision, predictive ML models | Predictive analytics, NLP-based assistants |
| Number of Agents | 5 | 5 |
| Number of Environments | 5 | 5 |
| Number of PEAS Combinations | 10 | 10 |
| Main Sensors | Soil, moisture, camera, weather sensors | Scores, activity logs, response data |
| Main Actuators | Pumps, valves, alerts, sprayers | Feedback, explanations, recommendations |
| Major Benefit | Improved yield and resource efficiency | Personalized, faster learning support |
| Major Risk | Incorrect recommendation causing crop loss | Overdependence, reduced human interaction |
| Privacy Concern | Farm and location data ownership | Student activity and performance data |
| Bias Concern | Region/crop-specific model accuracy | Fairness across different learner types |
| Major Ethical Issue | Data ownership and human oversight | Surveillance and academic integrity |
| Major Legal Issue | Liability for wrong recommendations | Student data protection and consent |
| Human Supervision | Farmer verifies and applies advice | Teacher guides and validates outcomes |

## 6. Common Ethical and Social Issues of AI

A few issues show up in both applications, just in slightly different forms:

- Privacy: AI systems may collect and store personal or sensitive data.
- Bias: Biased data can lead to unfair or inaccurate AI decisions.
- Transparency: Users should have a basic understanding of how AI is being used.
- Accountability: There should be clear responsibility when an AI system makes a harmful decision.
- Security: AI systems and the data they use should be protected from unauthorized access.
- Human Oversight: Important decisions should include human judgement instead of depending completely on AI.

## 7. The Role of Human Judgment in AI-Based Decision Making

- AI as a Support Tool: AI should assist humans rather than completely replace them.
- In Smart Agriculture: Farmers should consider AI recommendations along with their own experience and local conditions.
- In Education: Teachers should remain involved in important decisions about student learning and performance.
- Human Responsibility: Humans should remain responsible for important decisions made with the help of AI.
- Accountability: Clear responsibility is needed when an AI system makes an incorrect or harmful decision.
- Human Oversight: AI systems should be regularly monitored to ensure that their decisions are reliable and fair.

## 8. AI in the Current Scenario

- Smart Agriculture: AI is used for crop monitoring, disease detection, smart irrigation and yield prediction.
- Precision Farming: AI can help farmers use water, fertilizers and other resources more efficiently.
- AI in Education: AI is used for personalized learning, automated assessment and student performance analysis.
- AI Educational Assistants: AI tools can provide basic academic support and answer routine student questions.
- Generative AI: Tools such as generative AI can help students with explanations, ideas and learning support, but they should be used responsibly.
- Future Use: AI is expected to become more useful in agriculture and education as the technology improves.

## 9. Overall Conclusion

- AI Applications: Smart Agriculture and AI in Education show how AI can be used to solve practical problems.
- PEAS Analysis: The PEAS framework helps describe the agents, environments, inputs, outputs and performance measures of AI systems.
- Benefits: AI can improve decision-making, save resources and provide personalized support.
- Challenges: Privacy, bias, security, cost and overdependence on AI are important concerns.
- Ethical Responsibility: AI systems should be designed and used fairly and responsibly.

- Human Role: Human judgement and supervision should remain important when AI is used for important decisions.